

## f. Generate Shortage Order

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| <b>Textual Description</b> | When a product's inventory drops below its defined minimum threshold, the system automatically identifies the cheapest available supplier for that product and generates a new purchase order.                            |
| <b>List of Actors</b>      | Inventory Manager (Triggering the extension during an inventory update).                                                                                                                                                  |
| <b>Pre-conditions</b>      | <ul style="list-style-type: none"><li>• Products are defined in the system with a minimum stock threshold.</li><li>• Suppliers and active agreements containing pricing for these products exist in the system.</li></ul> |
| <b>Post-conditions</b>     | <ul style="list-style-type: none"><li>• A new shortage order is created and saved in the system from the supplier offering the lowest total price.</li></ul>                                                              |

### Main Success Scenario

1. The Inventory Manager inputs an inventory reduction update for a specific item.
2. The system saves the new quantity and detects that the total stock for the item has dropped below the minimum threshold.
3. The system retrieves all suppliers that have an active agreement to supply this specific item.
4. The system calculates and selects the cheapest supplier (factoring in any quantity-based discount brackets).
5. The system generates and saves a new purchase order from the selected supplier to cover the shortage.
6. The system displays an alert message to the Inventory Manager indicating that an automatic shortage order was created.

### Alternatives / Extensions

**2a. Inventory remains above the minimum threshold:**

1. The system verifies the updated quantity is still above the minimum threshold.
2. The use case terminates (no order is generated).

**3a. No supplier is defined for the missing item:**

1. The system detects that no supplier agreements contain the current item's catalog number.
2. The system logs an error and displays a warning message to the Inventory Manager: *"Automatic Error - No supplier linked to the missing item."*
3. The use case terminates without generating an order.